$$(r^2 + l^2)^{\frac{3}{2}} \approx r^3 \qquad (3.21)$$

Therefore, using equation (3.21) in equation (3.20), we get

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{p_m}{r^3}\hat{i}$$

Since $p_m \hat{i} = \vec{p}_m$, In general, the magnetic field at equatorial point is given by

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{\vec{p}_m}{r^3} \qquad (3.22)$$

Note that magnitude of B_{axial} is twice that of magnitude of $B_{equatorial}$ and the direction of B_{axial} and $B_{equatorial}$ are opposite.

EXAMPLE 3.6

A short bar magnet has a magnetic moment of 0.5 J T^{-1}. Calculate magnitude and direction of the magnetic field produced by the bar magnet which is kept at a distance of 0.1 m from the center of the bar magnet along (a) axial line of the bar magnet and (b) normal bisector of the bar magnet.

Solution

Given magnetic moment 0.5 J T^{-1} and distance r = 0.1 m

(a) When the point lies on the axial line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2p_m}{r^3}\right)\hat{i}$$

$$\vec{B}_{axial} = 10^{-7} \times \left(\frac{2\times 0.5}{(0.1)^3}\right)\hat{i} = 1\times 10^{-4} T\,\hat{i}$$

Hence, the magnitude of the magnetic field along axial is B_{axial} = 1 x 10^{-4} T and direction is towards South to North.

(b) When the point lies on the normal bisector (equatorial) line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{equatorial} = -\frac{\mu_\circ}{4\pi}\frac{p_m}{r^3}\hat{i}$$

$$\vec{B}_{equatorial} = -10^{-7}\left(\frac{0.5}{(0.1)^3}\right)\hat{i} = -0.5\times 10^{-4} T\,\hat{i}$$

Hence, the magnitude of the magnetic field along axial is $B_{equatorial}$ = 0.5 x 10^{-4} T and direction is towards North to South.

Note that magnitude of B_{axial} is twice that of magnitude of $B_{equatorial}$ and the direction of B_{axial} and $B_{equatorial}$ are opposite.

3.3 TORQUE ACTING ON A BAR MAGNET IN UNIFORM MAGNETIC FIELD

Consider a magnet of length $2l$ of pole strength q_m kept in a uniform magnetic field $\vec{B}$ as shown in Figure 3.19. Each pole experiences a force of magnitude $q_m B$ but acts in opposite direction. Therefore, the net force exerted on the magnet is zero, so that there is no translatory motion. These two forces constitute a couple (about midpoint of bar magnet) which will rotate and try to align in the direction of the magnetic field $\vec{B}$.

The force experienced by north pole,

$$\vec{F}_N = q_m \vec{B} \qquad (3.23)$$

The force experienced by south pole,

$$\vec{F}_S = -q_m \vec{B} \qquad (3.24)$$

Adding equations (3.23) and (3.24), we get the net force acting on the dipole as

$$\vec{F} = \vec{F}_N + \vec{F}_S = \vec{0}$$

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